# HOMEBUILT

YOUR DIY HOME IMPROVEMENT PARTNER




Krishna Bhatt

Diamond Challenge World Summit [2025]

Team 69973801

**Problem for Homeowners:** **Remodeling outdoor spaces is stressful for homeowners and hiring contractors is costly and uncertain.**




*Data from HomeBuilt Pain-Probing Survey (n=200)

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**Problem for Suppliers:** **Small hardware stores struggle to compete with larger chains, leaving them with growing unsold stock.**




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# The Solution: HomeBuilt Build Packs

## Mobile App



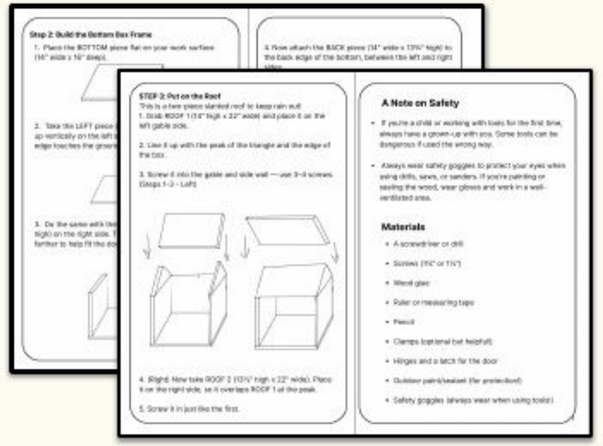







## Build Pack

Pet Houses, Gazebos, Garden Beds, Fencing, Tiling, and more!




Comprehensive Instruction Manual + 24/7 Support

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# The Solution: From Idea to Creation

🛠️ *Choose Renovation Type* → **AR Visualization**

✏️ *Edit and Customize* → **Benji AI Assistance**

📐 *Blueprint Generation* → **Benji SUPER Model**

📦 *Materials Packaged* → **Warehouse Dispatch**

🚚 *Shipping* → **24/7 Live Technical Support**

(+ Human Review)



Aayush and Benji




Krishna Bha

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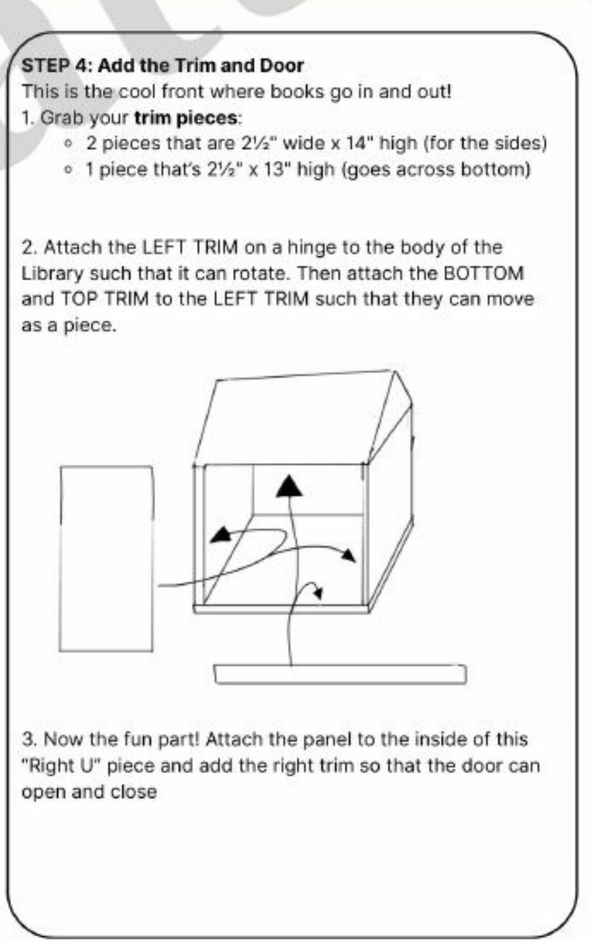
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**STEP 4: Add the Trim and Door**

This is the cool front where books go in and out!

1. Grab your **trim pieces**:
   - 2 pieces that are 2½" wide x 14" high (for the sides)
   - 1 piece that's 2½" x 13" high (goes across bottom)

2. Attach the LEFT TRIM on a hinge to the body of the Library such that it can rotate. Then attach the BOTTOM and TOP TRIM to the LEFT TRIM such that they can move as a piece.



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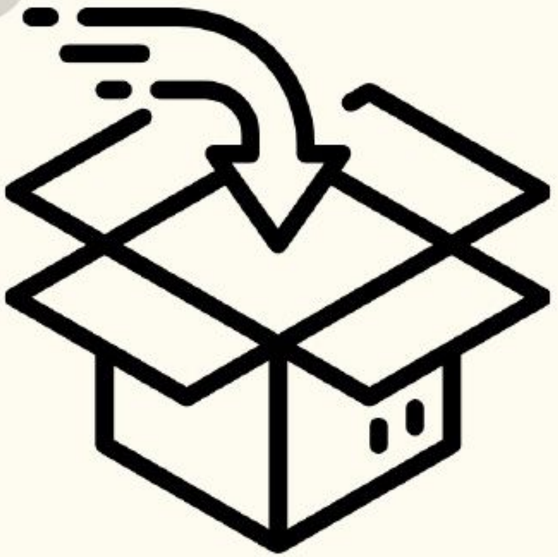
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# Your Home Improvement Assistant: Benji AI




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# Your Home Improvement Assistant: Benji AI

01 - Assists User in Designing

02 - Finds Required Materials

03 - Generates Instructions



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**Multimodal Self-Trained Large Language and Vision Assistant Model**

LLM + ViT + RAG

Large Language Model

Vision Transformer

Retrieval Augmented Generation

# Unique Value Proposition

## Homeowners

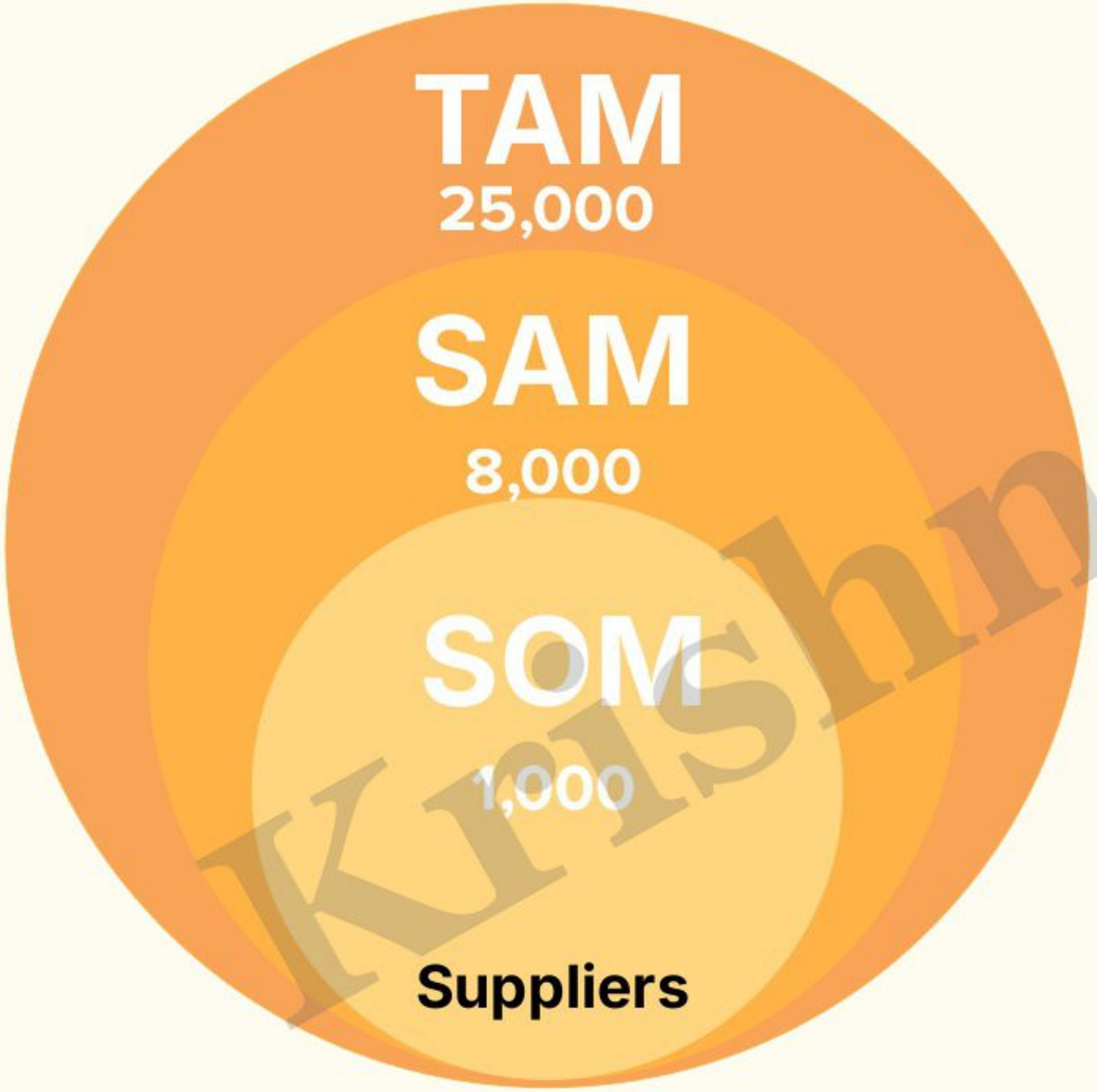
1. Customizability
2. Affordable Pricing
3. Fast Timelines
4. Trust and Quality
5. Enjoyable Installation

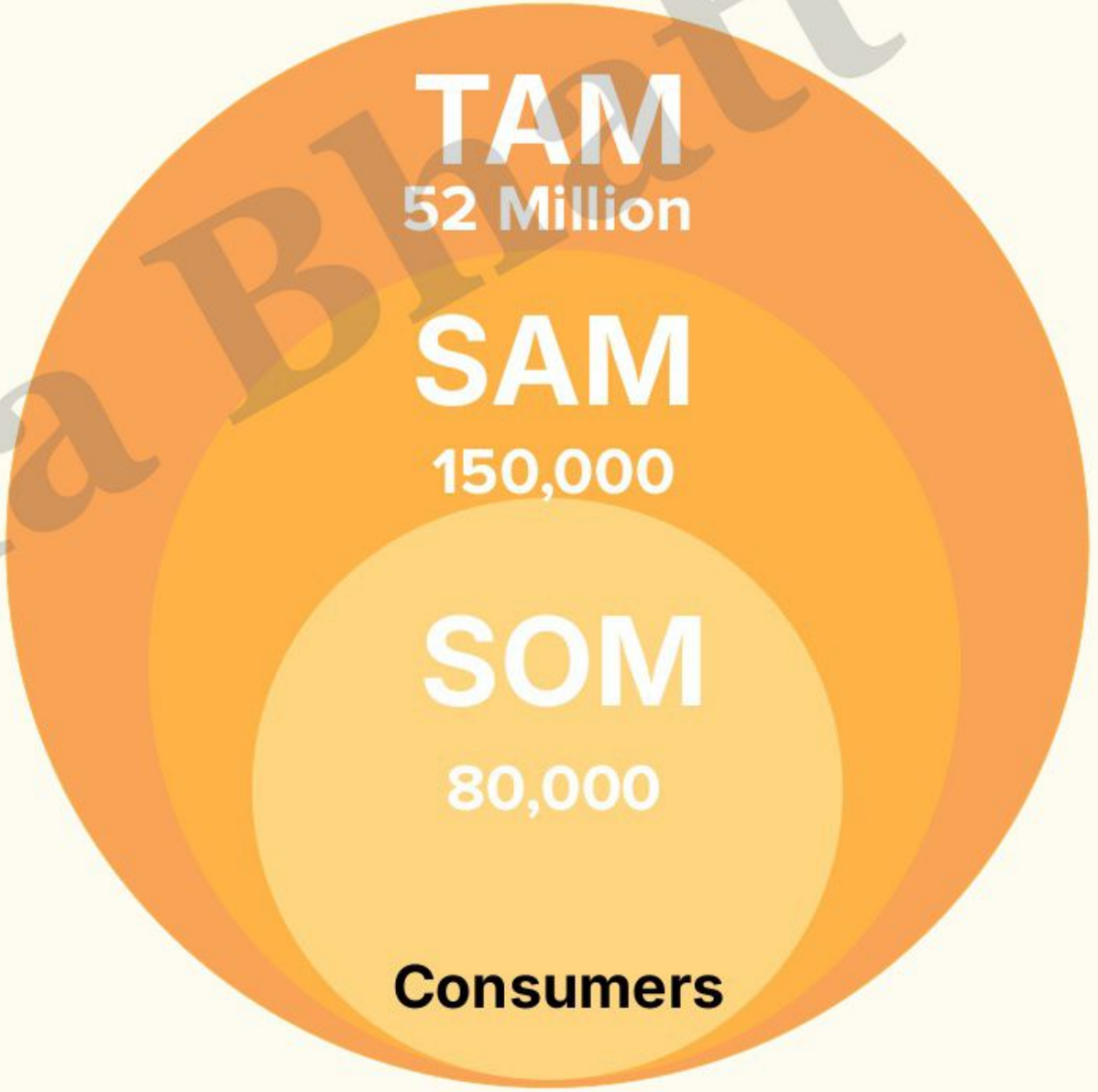
## Suppliers

1. Outlet for Excess Stock
2. Passive Revenue
3. Amplified Visibility

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# Market Opportunity




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# Revenue Model: Cost-Efficient Procurement Platform (CEPP)

- **Supplier Fees** ($40-80/mo)
- Excess Stock / Material

- 24/7 Technical Support
- Custom Build Packs

**Suppliers**




**Home-owners**

- **Flat Fee for Stock** (48%)
- Advertising Exposure

- **Build Pack Sale** ($300 - $6k)
- **Premium Subscription** ($49.99/yr)
- Platform Engagement

**Costs:**
Marketing, Remainder Stock, Cloud Computing, Warehousing

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# Competitive Advantage: Why CEPP Wins

| | HomeBuilt | Remodel AI | Contractors | Prebuilds |
|---|---|---|---|---|
| AR Visualization | ✓ | ✓ | ✗ | ✗ |
| Customizability | ✓ | ✓ | ✓ | ✗ |
| Customer Support | ✓ | ✗ | ✓ | ✓ |
| Affordable Pricing | ✓ | ✗ | ✗ | ✓ |
| Quality Material and Service | ✓ | ✓ | ✓/✗ | ✗ |

52% Higher Margins (Reduced COGS)

Compound Network Effects

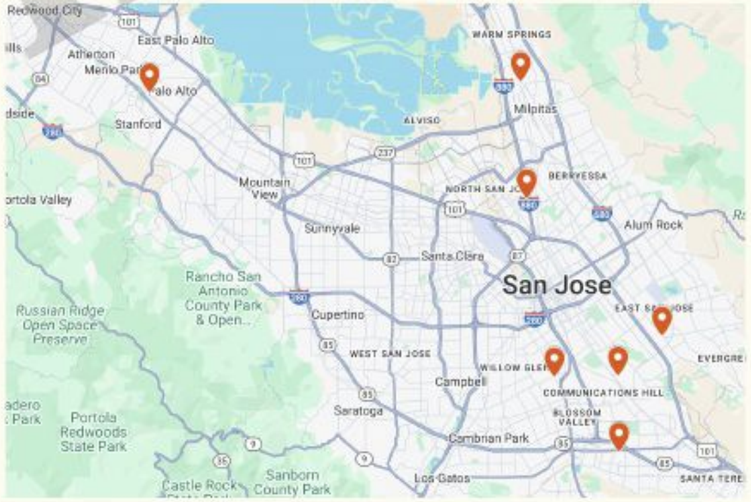
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# Go-to-Market: Current Progress

## FY2024 – FY2025

- **Completed HomeBuilt app development** (AR Visualization, Benji AI, Benji SUPER, E-commerce)
- **Secured 7 Affiliate Suppliers** for product launch with under CEPP model
- Launched **HomeBuilt Beta Testing Program**.






**Secured Affiliates:** Evergreen Supply, Ace Hardware Willow Glen, Hassett Hardware, Milpitas Material Co., Valley Tool & Supply Inc., Payless Hardware, Ace San Jose

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# Long-term Roadmap: 5 Year Timeline

## Year 1

- Launch full Bay Area pilot
- Secure 15+ Affiliated Suppliers
- Establish operations warehouse
- 100 Sales

## Year 2

- Expand to California metros (LA, San Diego, Sacramento)
- 225 Affiliated Suppliers
- 1214 Sales
- Break-even

## Year 3

- Expand to Southwest metros (Austin, Phoenix)
- 322 Affiliated Suppliers
- 5264 Sales,
- __ suppliers

## Year 4

- Secure 685 suppliers
- Official national expansion
- 3 Additional operational centers
- 12332 Sales

## Year 5

- 1000 Affiliated Suppliers
- 49,206 Sales
- Industry dominance

**Initial Aggressive Marketing Campaign**: HomeBuilt Project challenges, Creator Collaborations, etc.

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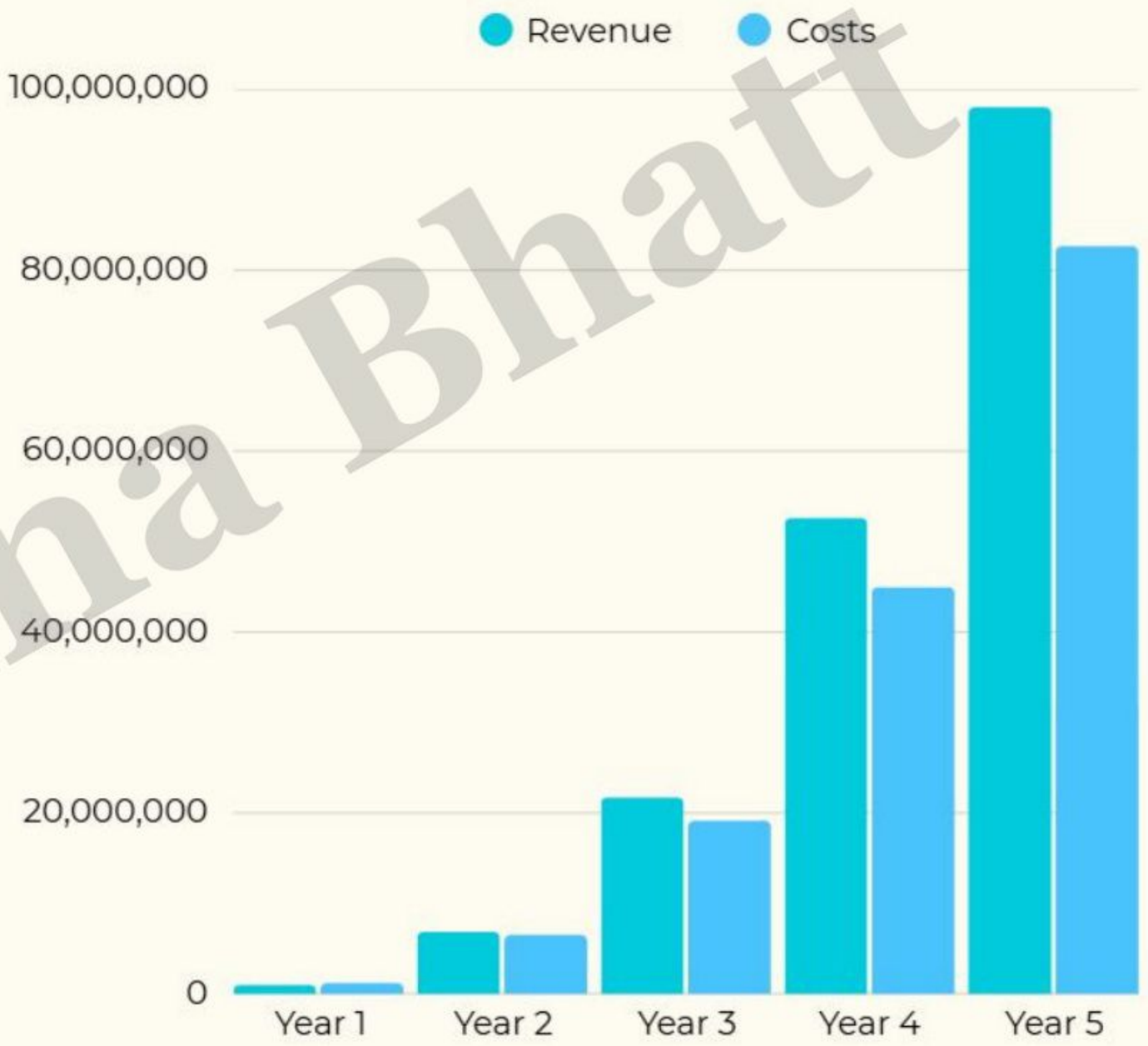
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# Financial Projections

**Average Revenue Per Customer**
$1800

**Average Revenue Per Supplier**
$600

**Customer Acquisition Costs**
$13



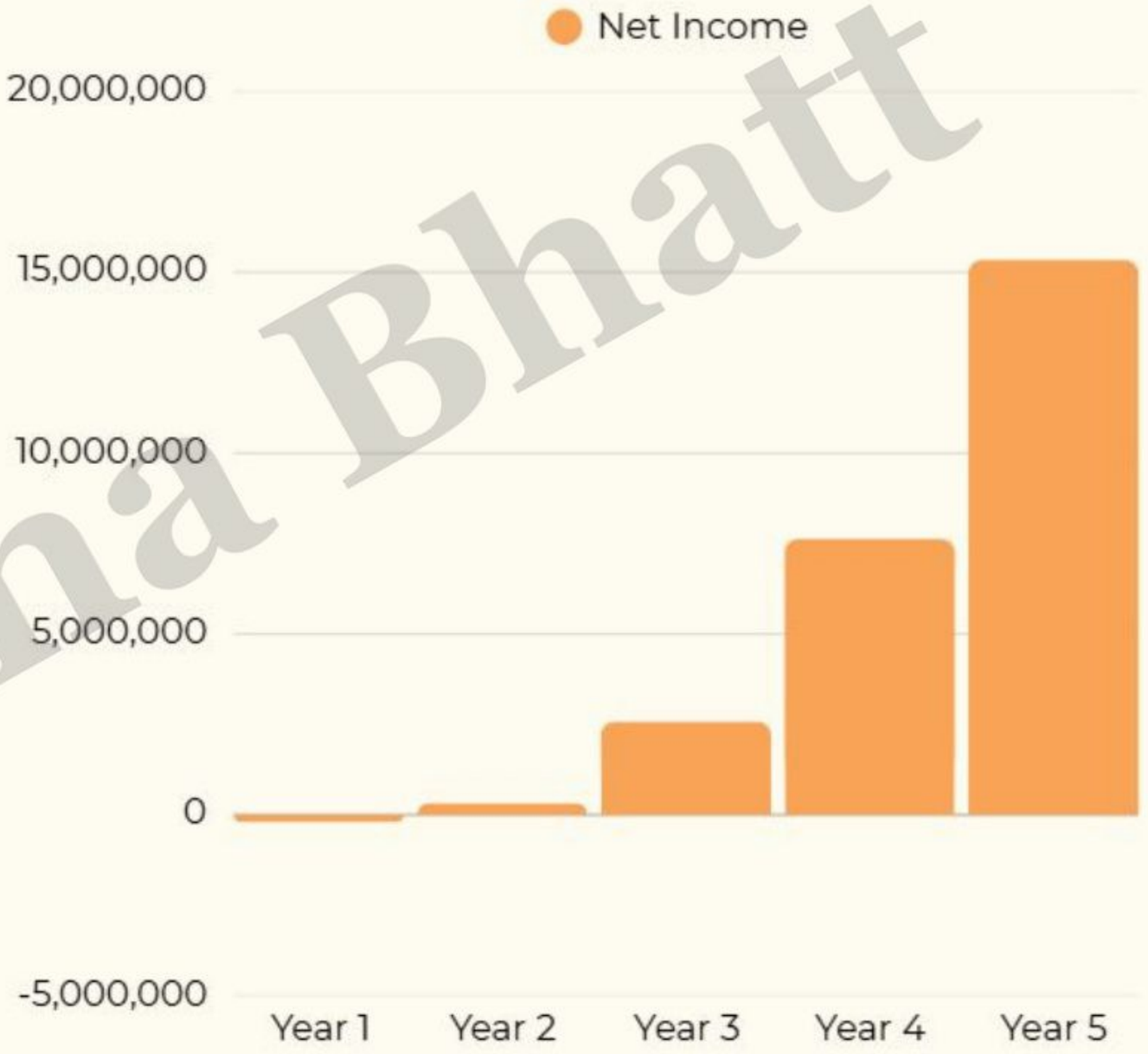

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# Financial Projections

**$26,655,865**
**Total Profit in 5 Years**

**$208,525**
**Loss in Year 1**

**Break Even by Year 2**




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# Testimonials: Beta Users and Suppliers




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# JOIN US

*TO REINVENT HOME IMPROVEMENT*

**HOMEBUILT**

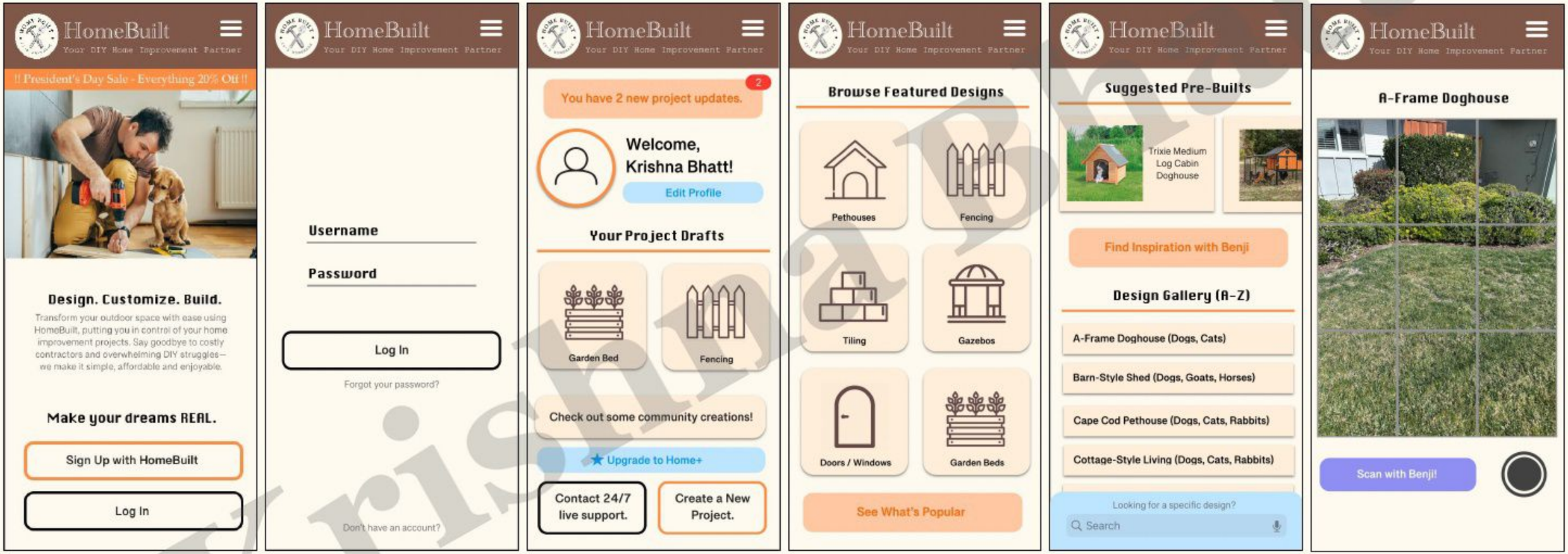
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# Appendix

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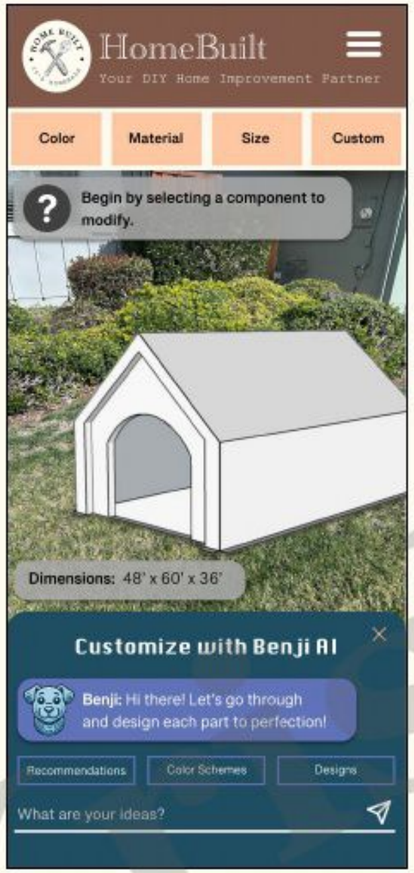
# Appendix: Mobile User Experience




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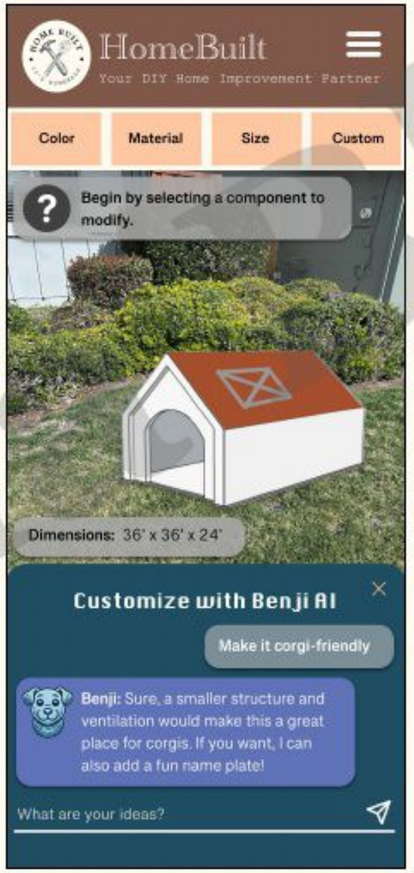
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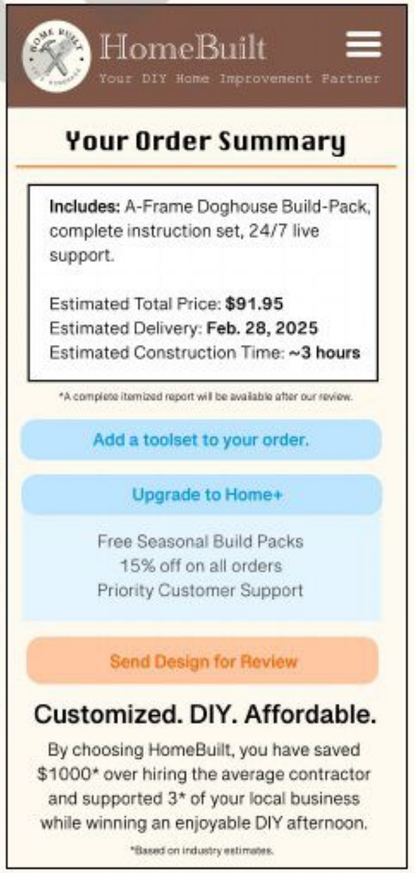












# Appendix: Sample Instructions (Benji SUPER V1.1)

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HOME BUILT
IT'S HOMEMADE

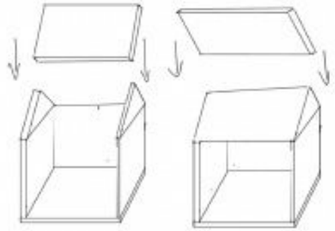
A comprehensive guide to build a
**Little Free Library**

HOMEBUILT
YOUR DIY HOME IMPROVEMENT PARTNER

**STEP 3: Put on the Roof**
This is a two-piece slanted roof to keep rain out!
1. Grab ROOF 1 (14" high x 22" wide) and place it on the left gable side.

2. Line it up with the peak of the triangle and the edge of the box.

3. Screw it into the gable and side wall — use 3–4 screws. (Steps 1-3 - Left)



4. (Right) Now take ROOF 2 (13½" high x 22" wide). Place it on the right side, so it overlaps ROOF 1 at the peak.

5. Screw it in just like the first.

## A Note on Safety

- If you're a child or working with tools for the first time, always have a grown-up with you. Some tools can be dangerous if used the wrong way.
- Always wear safety goggles to protect your eyes when using drills, saws, or sanders. If you're painting or sealing the wood, wear gloves and work in a well-ventilated area.

## Materials

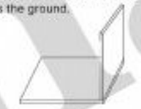
- A screwdriver or drill
- Screws (1¼" or 1½")
- Wood glue
- Ruler or measuring tape
- Pencil
- Clamps (optional but helpful!)
- Hinges and a latch for the door
- Outdoor paint/sealant (for protection!)
- Safety goggles (always wear when using tools!)

**Step 2: Build the Bottom Box Frame**

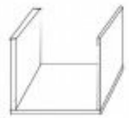
1. Place the BOTTOM piece flat on your work surface (14" wide x 16" deep).



2. Take the LEFT piece (16" wide x 14" high) and stand it up vertically on the left side of the bottom. The long edge touches the ground.



3. Do the same with the RIGHT piece (17" wide x 14" high) on the right side. This one will stick out just 1" farther to help fit the door later.



4. Now attach the BACK piece (14" wide x 13¾" high) to the back edge of the bottom, between the left and right sides.



Fasten them together:
- Put wood glue on the edges first.
- Use 2–3 screws on each connection to hold the back and sides to the bottom.

**STEP 3: Add the Gables**

These are the triangle pieces that support the roof.

1. Get your two **GABLE triangles** (each 15" wide at the bottom).



2. Put one triangle on top of the LEFT wall, and the other on the RIGHT wall.



3. Line them up so the bottoms match the wall tops exactly.

4. Screw each gable into the side walls with 2–3 screws per triangle.



TIP: Use a Clamp or a Helper!
When you place the gables on top of the side walls, it can be tricky to hold them steady while you screw them in. Use a small clamp to gently hold the triangle in place or ask a helper to hold it while you screw it down.

**Step 1: Cutting all pieces**

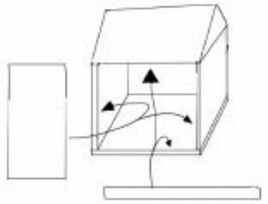
Safety first! Ensure you have the proper safety equipment, such as goggles and gloves, before beginning any cutting. Use a saw appropriate for the material, such as a miter saw or circular saw, to cut the pieces to the specified dimensions.

| | |
|---|---|
| Back | 14" W × 13½" H |
| Left Side | 16" W × 14" H |
| Right Side | 17" W × 14" H |
| Bottom | 14" W × 16" D |
| Roof 1 | 14" H × 22" W |
| Roof 2 | 13½" H × 22" W |
| Gables | 15" W (triangle) |
| Trim | 2½" W x various heights |
| Door | 12" H x 2½" W (with 8" tall side frames) |

Carefully measure and mark the dimensions on the materials before cutting.
Ensure all pieces are cut accurately to the specified sizes.

**STEP 4: Add the Trim and Door**
This is the cool front where books go in and out!
1. Grab your **trim pieces**:
   - 2 pieces that are 2½" wide x 14" high (for the sides)
   - 1 piece that's 2½" x 13" high (goes across bottom)

2. Attach the LEFT TRIM on a hinge to the body of the Library such that it can rotate. Then attach the BOTTOM and TOP TRIM to the LEFT TRIM such that they can move as a piece.



3. Now the fun part! Attach the panel to the inside of this "Right U" piece and add the right trim so that the door can open and close

4. Now attach the latch and you have finished!



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# Our Impact

**Fun, quality projects, free of the stresses of contractors**

**18%**
decrease in suppliers' costs.

**25%**
more affordable than contractors

**80%**
of homeowners believe they would greatly benefit with HomeBuilt

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# Meet the Team



## Archit Deepak

Technology

2x DECA ICDC Qualifier + 5x DECA Award Winner
2nd in the Western Region for Virtual Business Challenge
ML Experience w/ Internships @ MIT and AMD




## Krishna Bhatt

Strategy

1st Engineering @ USA Science Fair
2x TEDx Keynote Speaker
ML Research @ UC Berkeley
Recognition from White House, IBM, San Jose City Council



## Aayush Hegde

Partnerships

3+ years in ENGin Outreach
Editor-in-chief for Youtube channels
9x VRC Award Winner
Middle School VEX V5RC Mentor

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# Appendix: Cash Flow

| | Unit | 2025 FY2025 | 2026 FY2026 | 2027 FY2027 | 2028 FY2028 | 2029 FY2029 |
|---|---|---|---|---|---|---|
| **Cash Flow Statement** | | | | | | |
| **Cash and cash equivalents, BoP** | **[USD]** | **30,000** | **(130,032)** | **254,015** | **2,828,248** | **10,457,322** |
| Net income | [USD] | (208,525) | 330,816 | 2,572,616 | 7,625,634 | 15,335,324 |
| Adjustments to net income | | | | | | |
| Depreciation and Amortization | [USD] | - | - | - | - | - |
| Other operating expense | [USD] | - | - | - | - | - |
| Deferred income taxes | [USD] | (52,131) | 52,131 | - | - | - |
| Changes in operating assets and liabilities | | | | | | |
| Inventories | [USD] | - | - | - | - | - |
| Accounts receivable | [USD] | - | - | - | - | - |
| Accounts payable | [USD] | - | - | - | - | - |
| Accrued expenses | [USD] | - | - | - | - | - |
| Unearned revenue | [USD] | 625 | 1,100 | 1,617 | 3,440 | 4,735 |
| **Net cash from operating activities** | **[USD]** | **(260,032)** | **384,047** | **2,574,233** | **7,629,074** | **15,340,059** |
| Capex | [USD] | - | - | - | - | - |
| **Net cash from investment activities** | **[USD]** | **-** | **-** | **-** | **-** | **-** |
| Equity Issuance | [USD] | 100,000 | - | - | - | - |
| **Net cash from financing activities** | **[USD]** | **100,000** | **-** | **-** | **-** | **-** |
| **Net cash increase / (decrease) for the peric** | **[USD]** | **(160,032)** | **384,047** | **2,574,233** | **7,629,074** | **15,340,059** |
| **Cash and cash equivalents, End of Period** | **[USD]** | **(130,032)** | **254,015** | **2,828,248** | **10,457,322** | **25,797,381** |

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# Appendix: Profit and Loss I.

| | Unit | 2025 FY2025 | 2026 FY2026 | 2027 FY2027 | 2028 FY2028 | 2029 FY2029 |
|---|---|---|---|---|---|---|
| **P&L** | | | | | | |
| Build Box Margins | [USD] | 235,908 | 1,596,672 | 5,063,636 | 12,258,032 | 22,837,542 |
| Subscription Revenue | [USD] | 317 | 2,625 | 5,804 | 11,454 | 20,517 |
| **Revenues** | **[USD]** | **236,225** | **1,599,297** | **5,069,440** | **12,269,486** | **22,858,059** |
| Payment fees | [USD] | (5,906) | (39,982) | (126,736) | (306,737) | (571,451) |
| Hosting | [USD] | (335) | (2,439) | (5,249) | (10,575) | (19,461) |
| Customer Support (outsourcing) | [USD] | (386) | (2,350) | (6,654) | (14,764) | (26,420) |
| **Gross Profit** | **[USD]** | **229,598** | **1,554,526** | **4,930,801** | **11,937,410** | **22,240,726** |
| *% Net Revenue* | *[%]* | *97.2%* | *97.2%* | *97.3%* | *97.3%* | *97.3%* |
| Paid Marketing | [USD] | (57,165) | (239,386) | (550,401) | (1,072,435) | (1,625,395) |
| Management | [USD] | - | - | - | - | - |
| Engineering | [USD] | - | (21,000) | (78,000) | (108,000) | (108,000) |
| Marketing | [USD] | - | (63,000) | (108,000) | (108,000) | (108,000) |
| Customer Support | [USD] | - | - | - | - | - |
| Operations | [USD] | (13,090) | (48,620) | (119,680) | (157,080) | (157,080) |
| Payroll | [USD] | (13,090) | (132,620) | (305,680) | (373,080) | (373,080) |

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# Appendix: Profit and Loss II.

| | Unit | 2025 FY2025 | 2026 FY2026 | 2027 FY2027 | 2028 FY2028 | 2029 FY2029 |
|---|---|---|---|---|---|---|
| Material, equipment | [USD] | - | - | - | - | - |
| Software, tools, phone | [USD] | - | - | - | - | - |
| Travel | [USD] | - | - | - | - | - |
| Team expenses | [USD] | - | - | - | - | - |
| Other recurring expenses | [USD] | (420,000) | (769,000) | (858,950) | (959,853) | (1,073,096) |
| **EBITDA** | **[USD]** | **(260,657)** | **413,520** | **3,215,770** | **9,532,042** | **19,169,155** |
| *% Net Revenue* | *[%]* | *(110.3)%* | *25.9%* | *63.4%* | *77.7%* | *83.9%* |
| Depreciation & Amortization | [USD] | - | - | - | - | - |
| **EBIT** | **[USD]** | **(260,657)** | **413,520** | **3,215,770** | **9,532,042** | **19,169,155** |
| Net interest | [USD] | - | - | - | - | - |
| **EBT** | **[USD]** | **(260,657)** | **413,520** | **3,215,770** | **9,532,042** | **19,169,155** |
| Corporate taxes | [USD] | 52,131 | (82,704) | (643,154) | (1,906,408) | (3,833,831) |
| **Net Income** | **[USD]** | **(208,525)** | **330,816** | **2,572,616** | **7,625,634** | **15,335,324** |

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# Competitive Advantage: Why CEPP Wins

| | HomeBuilt | Remodel AI | Contractors | Prebuilds |
|---|---|---|---|---|
| AR Visualization |  |  |  |  |
| Customizability |  |  |  |  |
| Customer Support |  |  |  |  |
| Affordable Pricing |  |  |  |  |
| Quality Material and Service |  |  |  |  |

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52% Higher Margins (Reduced COGS)

*Over primary industry servers

Dual-Faced Monetization

Great Affiliation Incentive

Repeatable Across Metros

Compound Network Effects

# Why CEPP Wins

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